

FACE PUZZLE GAME

Webcam Capture · Sliding Tile Puzzle · Day 20 Build Log

What this is

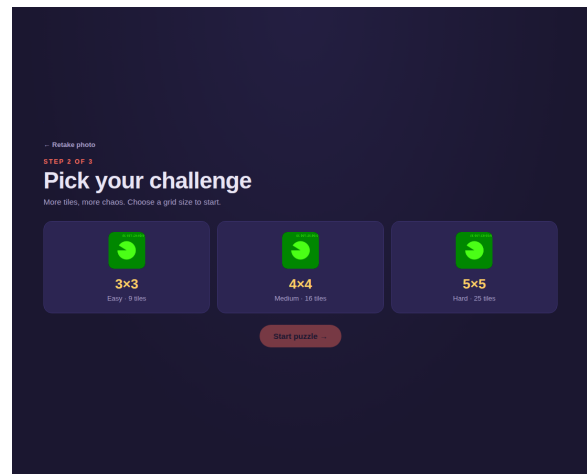
A browser-based puzzle game: capture a selfie via webcam, slice it into a sliding tile grid, and reassemble it by dragging tiles. Includes a live timer, move counter, and a local best-times leaderboard per difficulty level. Built as a single self-contained HTML file — no server or install required.

Gameplay Walkthrough



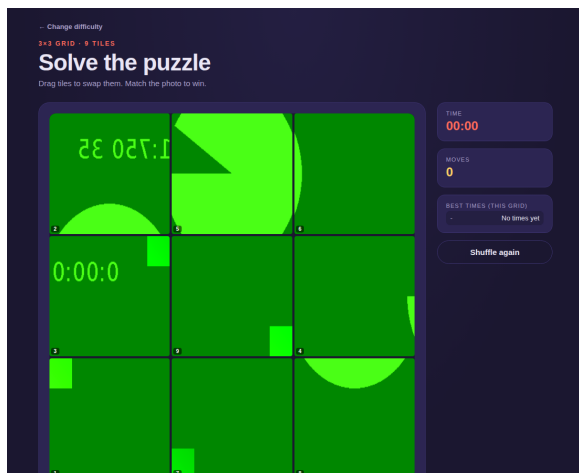
1. Capture screen

Webcam opens with live preview inside a photobooth-style frame.



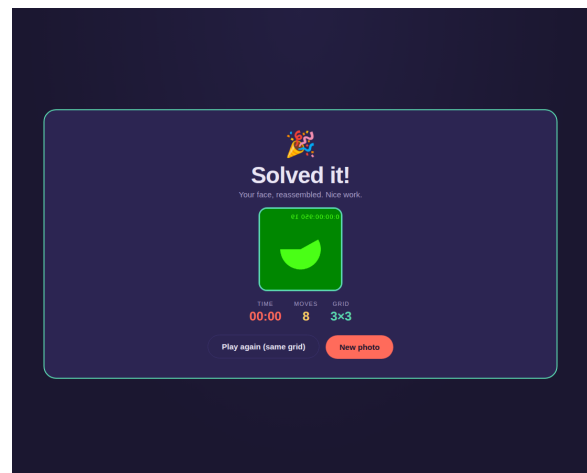
2. Difficulty select

Choose 3x3 (easy), 4x4 (medium), or 5x5 (hard) — each shows a live thumbnail of the captured photo.



3. Puzzle board

Tiles are sliced from the photo and shuffled. Sidebar tracks time, moves, and best times for this grid size.



4. Solved

Win screen confirms completion with final time, move count, and the reassembled photo.

Feature Checklist

Feature	Status
Webcam capture with live preview	Implemented
Photo retake before confirming	Implemented
3×3 / 4×4 / 5×5 difficulty selection	Implemented
Tile slicing from captured photo	Implemented
Drag-and-drop tile swapping	Implemented
Touch support (tap-to-swap)	Implemented
Live timer	Implemented
Move counter	Implemented
Local leaderboard (best 5 times per grid)	Implemented
Win detection + completion screen	Implemented
Shuffle-again / play-again / new-photo flows	Implemented

Key Learnings

- Tile slicing math (background-size + background-position percentages) needs the grid size baked into both axes correctly, or tiles show the wrong crop — verified by testing each grid size (3, 4, 5) independently.
- A naive shuffle can occasionally land on an already-solved board; explicitly checking for that and re-shuffling avoids a puzzle that's "won" before it starts.
- Drag-and-drop alone excludes touch devices, so a tap-to-select / tap-to-swap fallback was added for mobile.
- Browser storage (localStorage) is the simplest way to persist a leaderboard across sessions without a backend, but needs a try/catch since some browser contexts block it.
- Testing logic in isolation (shuffle validity, win detection, leaderboard sort/trim) caught edge cases before they became visible bugs in the UI.

Deliverable: single-file HTML application (face_puzzle_game.html), tested end-to-end with an automated browser session including a real solved-game run.